Data Semantics

Assignemnt 1 :

1. When is her birthday?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?birthDate
WHERE {
   dbr:Jennifer_Aniston dbo:birthDate ?birthDate .
}
```

birthDate
1969-02-11

2. Where was she born ?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?personName ?birthPlaceName
WHERE {
   dbr:Jennifer_Aniston rdfs:label ?personName .
   dbr:Jennifer_Aniston dbo:birthPlace ?birthPlace .
```

```
    ?birthPlace rdfs:label ?birthPlaceName .

    FILTER(LANG(?personName) = "en")
    FILTER(LANG(?birthPlaceName) = "en")
}
```

personName	birthPlaceName
"Jennifer Aniston"@en	"Los Angeles"@en

3. Who is recorded as Jennifer Aniston's spouse in DBpedia?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?personName
WHERE {
    {
        dbr:Jennifer_Aniston dbo:spouse ?person .
    }
    UNION
    {
        dbr:Jennifer_Aniston dbo:partner ?person .
    }

    ?person rdfs:label ?personName .
```

```
    FILTER(LANG(?personName) = "en")
}
```

Answer :

SPARQL | HTML5 table

personName
"Brad Pitt"@en
"Justin Theroux"@en

4. How many films did she star in?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT (COUNT(DISTINCT ?film) AS ?numberOfFilms)
WHERE {
    ?film dbo:starring dbr:Jennifer_Aniston .
    ?film a dbo:Film .
}
```

SPARQL | HTML5 table

numberOfFilms
38

5. How old were Jennifer Aniston's partners?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?personName ?birthDate
      (YEAR(NOW()) - YEAR(?birthDate) AS ?age)
WHERE {
   {
      dbr:Jennifer_Aniston dbo:spouse ?person .
   }
   UNION
   {
      dbr:Jennifer_Aniston dbo:partner ?person .
   }

   ?person dbo:birthDate ?birthDate .
   ?person rdfs:label ?personName .
```

```
    FILTER(LANG(?personName) = "en")
}
ORDER BY ?personName
```

personName	birthDate	age
"Brad Pitt"@en	1963-12-18	63
"Justin Theroux"@en	1971-08-10	55

6. Which television series starring Jennifer Aniston has the highest number of episodes?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?seriesName ?numberOfEpisodes
WHERE {
    ?series dbo:starring dbr:Jennifer_Aniston .
    ?series dbo:numberOfEpisodes ?numberOfEpisodes .
    ?series rdfs:label ?seriesName .

    FILTER(LANG(?seriesName) = "en")
}
ORDER BY DESC(?numberOfEpisodes)
LIMIT 1
```

SPARQL | HTML5 table

seriesName	numberOfEpisodes
"Friends"@en	"236"^^<http://www.w3.org/2001/XMLSchema#nonNegativeInteger>

7. What is the genre of the Friends series?

```
PREFIX dbr: <http://dbpedia.org/resource/>

PREFIX dbo: <http://dbpedia.org/ontology/>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>


SELECT DISTINCT ?genreName

WHERE {

   dbr:Friends dbo:genre ?genre .

   ?genre rdfs:label ?genreName .


   FILTER(LANG(?genreName) = "en")

}
```

genreName
"Sitcom"@en

8. Who are the main actors starred in Friends series?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?actorName
WHERE {
   dbr:Friends dbo:starring ?actor .
   ?actor rdfs:label ?actorName .

   FILTER(LANG(?actorName) = "en")
}
ORDER BY ?actorName
```

actorName
"Courteney Cox"@en
"David Schwimmer"@en
"Jennifer Aniston"@en
"Lisa Kudrow"@en
"Matt LeBlanc"@en
"Matthew Perry"@en

9. How many seasons does the Friends series have??

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>

SELECT ?numberOfSeasons
WHERE {
   dbr:Friends dbo:numberOfSeasons ?numberOfSeasons .
}
```

numberOfSeasons
"10"^^<http://www.w3.org/2001/XMLSchema#nonNegativeInteger>

9. whan was frineds series first released ?

```
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX dbo: <http://dbpedia.org/ontology/>


SELECT ?releaseDate
WHERE {
   dbr:Friends dbo:releaseDate ?releaseDate .
}
```

SPARQL | HTML5 table

releaseDate
1994-09-22